



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
27-06-2021

JEE(Main+Advanced) : NURTURE COURSE (PHASE : TEPS)

PAPER-2

Time : 3 Hours

Maximum Marks : 183

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **26** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

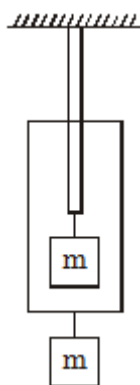
HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

PHYSICS

SEC-1 : (Maximum Marks: 21)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks : -1 In all other cases

1. The upper end of the steel rod of length 1 m and radius 0.5 cm is attached to a ceiling. Young's modulus of steel is 200 GPa. Find the elongation in the rod if we suspend two equal masses from the middle and lower end with the help of special suspension as shown. Mass $m = 400$ kg each, mass of the rod is negligible.



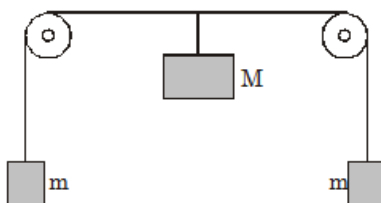
- (A) 0.38 mm (B) 0.56 mm (C) 0.19 mm (D) 0.13 mm

Space for Rough Work

2. A large plastic sphere floats on water with one sixteenth of its volume above the surface. Oil is poured over water till the sphere is just covered by oil. At this stage one half of the sphere is below water oil interface. What is density of oil?

(A) 940 kg/m^3 (B) 875 kg/m^3 (C) 700 kg/m^3 (D) 500 kg/m^3

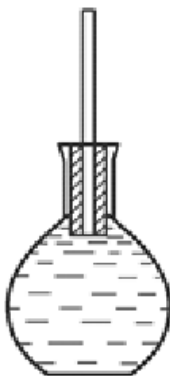
3. In the figure shown, the threads are massless and pulley is ideal. $M = m/4$. The mass M is released. What is the maximum distance descended by M if the distance between the small pulleys shown is $2a$. The mass M always remains on the perpendicular bisector of the line joining both the pulleys.



(A) $3a/2$ (B) $15a/8$ (C) $5a/6$ (D) $16a/63$

Space for Rough Work

4. A flask was tightly fitted with a cork fitted with a pipe. Kerosene was poured in the flask till the cork. By how many pascals will the pressure on the bottom of the flask change during heating of the kerosene by a temperature difference of 30°C if volume of the flask is 2 litres, its height $h = 20$ cm, cross section of the pipe is 2 cm^2 ? The coefficient of volume expansion of kerosene is $10^{-3} / ^\circ\text{C}$, its density before heating was $0.8 \times 10^3\text{ kg/m}^3$. Thermal expansion of the flask can be neglected.



(A) 1.97 kPa

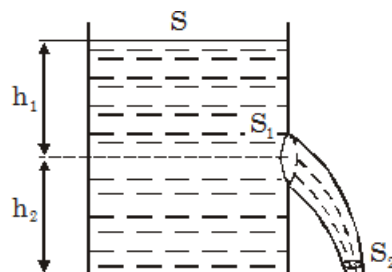
(B) 4.73 kPa

(C) 8.36 kPa

(D) 2.28 kPa

Space for Rough Work

5. A Cylindrical tank has a cross-sectional area $S = 1000 \text{ cm}^2$, stands motionless on a horizontal surface (figure). In its wall there is a hole, cross-sectional area of which $S_1 = 10 \text{ cm}^2$, located at a distance $h_1 = 80 \text{ cm}$ from the surface of water in the tank and the $h_2 = 180 \text{ cm}$ from the bottom. Find cross-sectional area S_2 of the jet emanating from the hole at the place of its falling on a horizontal surface.



- (A) 7.5 cm^2 (B) 5 cm^2 (C) $\frac{5}{\sqrt{13}} \text{ cm}^2$ (D) $\frac{20}{\sqrt{13}} \text{ cm}^2$
6. A mole of helium expands without any change in temperature of 300 K from an initial volume to a final volume absorbing a heat of $7.5 \ln 2 \text{ kJ}$ from the reservoir. Now the gas is made to undergo change from same initial volume to same final volume but without absorbing any heat from outside. What is the work done by the gas during this process? Take $R = 25/3 \text{ J/mol K}$
- (A) 1.25 kJ (B) 1.875 kJ (C) 2.815 kJ (D) 2.325 kJ
7. A small ball is falling in a liquid with a terminal speed of 8 m/s . If the same ball is dropped in a liquid flowing horizontally with a speed of 6 m/s , what is its terminal speed as seen from ground?
- (A) 10 m/s (B) 14 m/s (C) 12 m/s (D) 8 m/s

Space for Rough Work

SEC-2 : (Maximum Marks: 28)

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- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

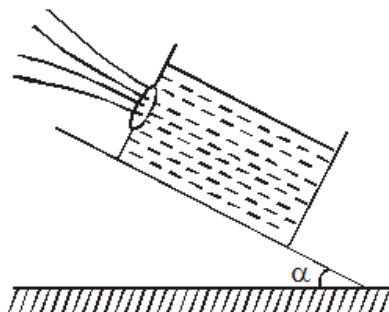
1. A capillary is dipped vertically in a soap solution. After that, at the top of the capillary, we cover the surface with a flat soap film (case 1). Now some air is pushed inside the air enclosed using a syringe so that a hemispherical bubble is formed at the top (case 2). After that, some air is pulled out from the air enclosed so that a hemispherical bubble is formed but with its curved surface inside the capillary (case 3). In the initial case, the height of liquid inside the capillary is 2 cm. The wetting is complete.
 - (A) In case 1, pressure of enclosed air is atmospheric
 - (B) In case 2, the height of liquid inside the capillary is 1 cm above the water surface
 - (C) In case 3, the height of liquid inside the capillary is 6 cm above the water surface
 - (D) Information is insufficient.

Space for Rough Work

2. Blocks of mass $m = 1 \text{ kg}$ and $M = 4 \text{ kg}$ have a spring in between. System was put on a smooth horizontal table. In the figure, the blocks are not connected to the spring. The spring is compressed by 10 cm in the shown position and the blocks are held in place by two stops. If the stop on the left is removed, m acquires speed of 8 m/s .



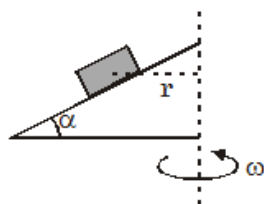
- (A) If instead of this, the stop on the right is removed, M acquires a speed of 4 m/s .
- (B) If instead of this, both the stops were removed simultaneously, both the masses acquire a momentum of $\frac{32}{\sqrt{5}} \text{ kg m/s}$.
- (C) If the spring was attached to both the blocks, on removal of left stop, the mass M acquires a maximum speed of $\frac{16}{\sqrt{5}} \text{ m/s}$.
- (D) If the spring was attached to both the blocks, on removal of left stop, the mass m acquires a maximum speed of 8 m/s .
3. A tank of negligible mass and of area $100A$ is full of water. It is sliding on a rough (coefficient of friction μ) inclined plane of angle α such the free surface of water is parallel to the inclined plane. There is a small hole of area A at the bottom of the rear wall from which water is being ejected with a variable relative velocity u . The height of water in the tank is y . At which of the combination of u , y , μ and α is it possible?



- (A) 10 m/s , 1 m , 0.1 and 37° (B) 10 m/s , $5/3 \text{ m}$, 0.1 and 53°
- (C) 10 m/s , 1 m , 0.2 and 60° (D) 20 m/s , 2 m , 0.1 and 45°

Space for Rough Work

4. A force is given by $x^2y^2 \hat{i} + x^2y^2 \hat{j}$. A particle of mass 1 kg is taken from origin to (1,1) by two paths. (i) First along y axis, then along a straight line parallel to x axis. (ii) First along x axis, then along a straight line parallel to y axis.
- (A) The force is conservative in nature.
- (B) The force is non-conservative in nature
- (C) The total work done by the force along both the paths is equal
- (D) The total work done by the force along both the paths is not equal.
5. On an inclined plane with angle $\alpha = 22.5^\circ$ is a body as shown in the figure. The plane uniformly rotates around a vertical axis. The distance from the body to the axis of rotation is $r = 10$ cm. Find the correct combination(s) of friction coefficient μ and ω at which the body does not slide down the inclined plane? $\sin 22.5^\circ = 5/13$



- (A) $\mu = 0.5, \omega = 10\sqrt{\frac{2}{29}} \text{ rad/s}$
- (B) $\mu = 0.25, \omega = 5\sqrt{\frac{2}{3}} \text{ rad/s}$
- (C) $\mu = 1.0, \omega = 10\sqrt{\frac{7}{19}} \text{ rad/s}$
- (D) $\mu = 0.4, \omega = 10\sqrt{2} \text{ rad/s}$

Space for Rough Work

6. Portion of helium in a cyclic process initially adiabatically expands, while the gas temperature decreases from 500 K to 497 K, then compressed isobarically to the original volume, and finally heated isochorically to initial temperature.
- (A) The volume of the gas increases during the adiabatic process by approx. 1%
- (B) The efficiency of the thermodynamic cycle is 2 %
- (C) The minimum temperature during the cycle is 492.5 K approx..
- (D) The maximum temperature during the cycle is 500 K.
7. Lower end of a capillary tube of radius 10^{-3} m is dipped vertically into a liquid. Surface tension of liquid is 0.5 N m and specific gravity of liquid is 5. Contact angle between liquid and material of capillary tube is 120° . Choose the correct options (use $g = 10\text{m/s}^2$)
- (A) Maximum possible depression of liquid column in the capillary tube is 1 cm.
- (B) Maximum possible depression of liquid column in the capillary tube is 2 cm.
- (C) If the length of the capillary tube dipped inside liquid is half of the maximum possible depression of liquid column in the capillary tube, angle made by the liquid surface at the end of the capillary tube with the vertical, is $\cos^{-1} \left(-\frac{1}{4} \right)$.
- (D) If the length of the capillary tube dipped inside liquid is one third of the maximum possible depression of liquid column in the capillary tube, angle made by the liquid surface at the end of the capillary tube with the vertical, is $\cos^{-1} \left(-\frac{1}{6} \right)$.

Space for Rough Work

SEC-3 : (Maximum Marks: 12)

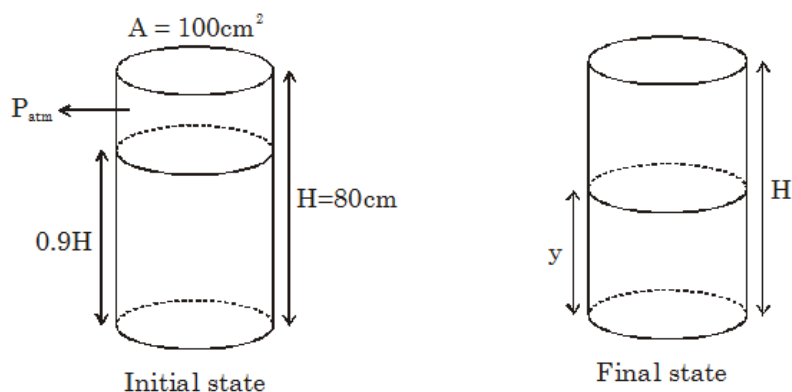
- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
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Full Marks : +3 If **ONLY** the correct option is chosen.

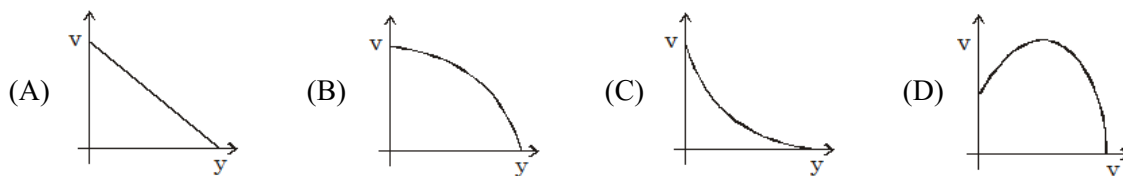
Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

I had purchased a water bottle which was sealed at the top. But i did not realize that it had a small hole at the bottom. Interestingly, water leaked out for some time and stopped leaking after that. Area of hole = 1 cm^2 . Assume that no air enters from hole into the bottle, $\rho_w = 1000 \text{ kg/m}^3$, $g = 10 \text{ m/s}^2$, $p_0 = 10^5 \text{ Pa}$.



1. What is the final height y of water in the bottle ?
 (A) Approx, 70 cm (B) Approx. 60 cm
 (C) Approx. 50 cm (D) Approx. 40 cm
2. The velocity of efflux from hole varies with y as :



Space for Rough Work

Paragraph for Questions 3 and 4

Billionaire Elon Musk wants to build a penthouse in space. He conceives a blackened spherical shell hovering in the outer space between earth and sun. His penthouse would be built inside this shell. For this purpose, he has hired a retired scientist from ISRO, Ms. Budhimati Thakur. Ms. Budhimati is an Indian, and she does not like this wasteful expenditure of foolish rich people while people here on the earth are starving to death. On the other hand, Mr. Musk is building castles in air (actually penthouses in space). He is busy thinking how the shell will maintain the temperature by being heated from Sun. The Black body radiations emitted by the shell will be received by earthlings and Aliens. They will all be able to see the glories of me and my family members.

Ms. Budhimati finally builds that required space station using special material that has high thermal conductivity, and negligible thermal expansion coefficient. But she uses a very thin material for the shell so that the shell will explode if the oxygen inside has a pressure more than 1.1 times the initial value. Initially, the shell is located at such a distance from the sun that the radiation incident on it is having an intensity of 1836 W/m^2 . Take $\sigma = \frac{17}{3} \times 10^{-8} \text{ W/m}^2\text{K}^4$, $b = 0.3 \text{ cm-K}$.

Disclaimer: This paragraph is a work of fiction and any resemblance to a person living or dead is purely coincidental.

3. What is the wavelength of most intense beam being radiated by the shell in steady state?
 (A) 7071 nm (B) 5000 nm (C) 10,000 nm (D) 20,000 nm
4. At what value of the incident intensity will the shell explode?
 (A) 2221 W/m^2 . (B) 2688 W/m^2 . (C) 2442 W/m^2 . (D) 1932 W/m^2 .

Space for Rough Work

CHEMISTRY

SEC-1 : (Maximum Marks: 21)

- This section contains **SEVEN (07)** questions.
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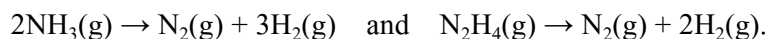
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

- For molecular formula C_4H_9NO , how many amide will be there which will not form H-bond (Only structural)
 (A) 1 (B) 2 (C) 3 (D) 4
- Which of the following is most basic in aqueous phase?
 (A) NH_3 (B) $MeNH_2$ (C) Me_2NH (D) Me_3N
- Identify set of nucleophiles :
 (A) $\overset{\ominus}{Cl}, \overset{\ominus}{OH}, \overset{\oplus}{R}, \overset{\square}{:CH_2}$
 (B) $CH_3 - \overset{\overset{O}{\parallel}}{C} - \overset{\ominus}{O}, \overset{\ominus}{N_3}, H^{\oplus}, \overset{\ominus}{SH}$
 (C) $CH_2 = CH_2, CH_3 - \overset{\cdot\cdot}{N}H_2, CH_3 - CH_2 - \overset{\cdot\cdot}{O}H, \text{Benzene ring}$
 (D) $\overset{\cdot\cdot}{H}^{\ominus}, \overset{\ominus}{:CN}, \overset{\oplus}{CH_3}, \text{p-Tolyl ring}$

Space for Rough Work

4. A mixture of $\text{NH}_3(\text{g})$ and $\text{N}_2\text{H}_4(\text{g})$ is placed in a sealed container at 300 K. The pressure within the container is 0.6 atm. The container is heated to 1000 K where the two gases undergo complete decomposition reactions



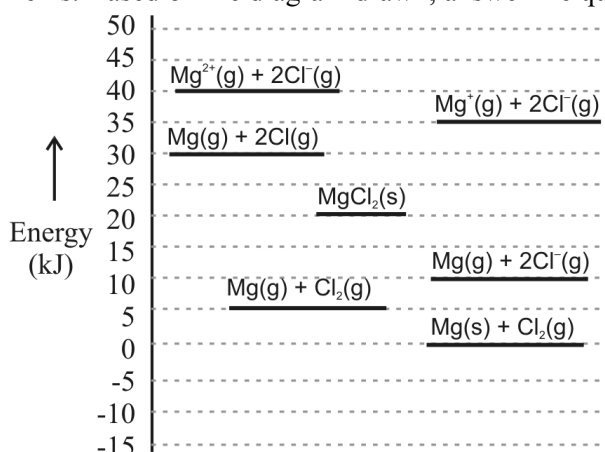
The pressure of the container at this stage becomes 4.8 atm. The mole per cent of $\text{NH}_3(\text{g})$ in the original mixture was

- (A) 40 (B) 50 (C) 60 (D) 70
5. When N_2O_5 is heated at temp. T, it dissociated as $\text{N}_2\text{O}_5 \rightleftharpoons \text{N}_2\text{O}_3 + \text{O}_2$, $K_c = 2.5$. At the same time N_2O_3 also decomposes as : $\text{N}_2\text{O}_3 \rightleftharpoons \text{N}_2\text{O} + \text{O}_2$. If initially 4 moles of N_2O_5 are taken in 1.0 litre flask and allowed to attain equilibrium, concentration of O_2 was formed to be 2.5M. Equilibrium concentration of N_2O is :-

- (A) 1.0 (B) 1.5 (C) 2.166 (D) 0.334

Space for Rough Work

6. Born Haber cycle helps to determine lattice enthalpy of ionic compound as well as electron gain enthalpy of different elements. Based on the diagram drawn, answer the questions given below.



What is the magnitude of lattice enthalpy of MgCl_2 ?

- (A) 20kJ (B) 40kJ (C) 15kJ (D) 40kJ
7. A planar sp^3 hybridised molecule having same surrounding atoms has 18% s-character in a bond. The % p-character in a lone pair of central atom is-
- (A) 18% (B) 64% (C) 82% (D) 68%

Space for Rough Work

SEC-2 : (Maximum Marks: 28)

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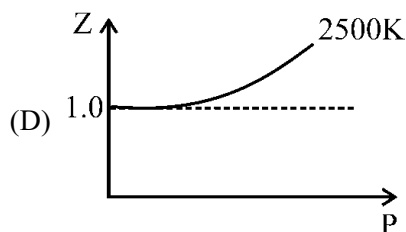
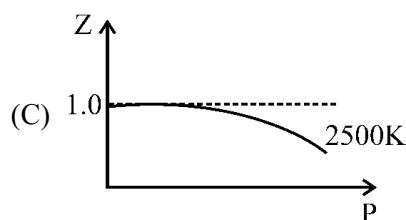
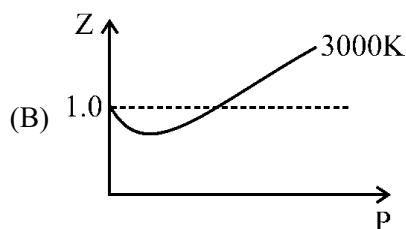
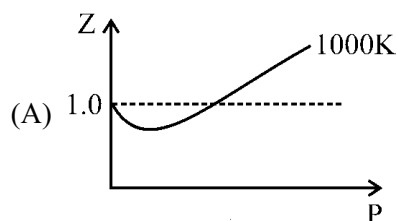
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. For a van der waal gas, $a = 4 \text{ atm-L}^2/\text{mol}^2$ and $b = 0.02 \text{ L/mol}$. Select the correct possible graph(s)
[Given : $R = 0.08 \text{ L-atm/mol-K}$]



Space for Rough Work

2. Vander waal's equation of state for real gases may be written as :

$$PV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

Select the correct statement(s) -

- (A) 'B' & 'C' are temperature dependent.
 (B) 'B' is temperature dependent while 'C' is temperature independent.
 (C) 'B' may be positive , negative or zero.
 (D) 'B' may be positive or zero but never negative.
3. Select the correct option(s) regarding XeO_3F_2
- (A) Max number of atoms in one plane = 4
 (B) Number of planes containing max atoms = 2
 (C) Three $2p\pi$ - $3d\pi$ bonds
 (D) $\text{O}=\text{Xe}=\text{O}$ bond angle is lesser than $\text{O}=\text{Xe}=\text{O}$ bond angle in XeO_2F_4
4. Hydrogen bonding influences structure of some compounds by linking atoms or groups in to larger structural units.

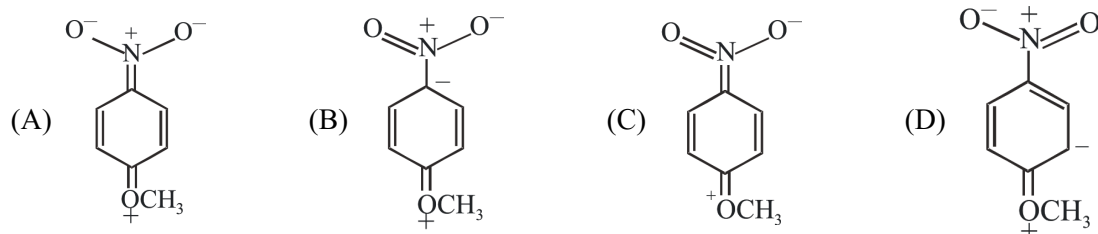
Select CORRECT combination for hydrogen bonded structure

- (A) Infinite linear chain : $\text{HCN}_{(s)}$
 (B) Infinite layers : $\text{H}_3\text{BO}_{3(s)}$
 (C) Three dimensional network structure : $\text{H}_2\text{O}_{(s)}$
 (D) Dimeric molecules : CH_3COOH in benzene

(A) A (B) B (C) C (D) D

Space for Rough Work

5. Which of the following pair of species have different hybridisation but equal bond angle :
- (A) I_3^- , ICl_2^- (B) CO_2 , XeF_2 (C) I_3^+ , Br_3^- (D) NH_3 , COCl_2
6. For an hypothetical atom, if the value of Azimuthal quantum number (ℓ) ranges from 0 to $n + 2$ and magnetic quantum number (m) ranges from $(-\ell - 1)$ to $(+\ell + 1)$ and having 3 spin quantum states in each orbital then which of the following statement is / are **INCORRECT** for $n = 1$ shell (n , ℓ , m are integers)
- (A) p-subshell can contain total 5 orbitals
- (B) Maximum number of electron accomodate in this shell is 48
- (C) Total 24 degeneracies are possible if it is a hydrogen like species
- (D) Maximum value of $n + \ell + m$ of an orbital in f -subshell will be 7
7. The correct resonating structure is -



Space for Rough Work

SEC-3 : (Maximum Marks: 12)

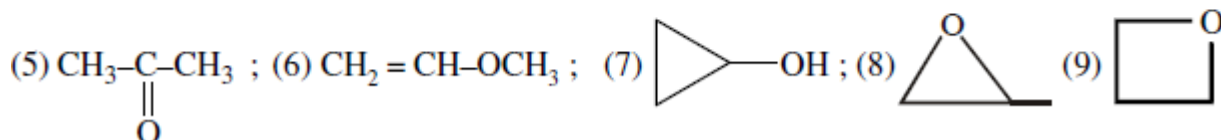
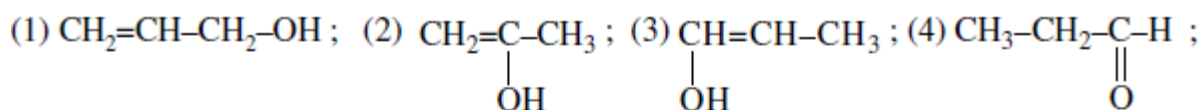
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Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Molecular formula C_3H_6O having following structures :



- Which type of isomerism is present between compound (1) and (4) :-
 (A) Chain isomer (B) Functional isomers
 (C) Position isomers (D) Metamers
- Which of the following set is not the example of positional isomerism :-
 (A) (4) & (5) (B) (2) & (3) (C) (1) & (2) (D) (1) & (3)

Space for Rough Work

Paragraph for Questions 3 and 4

A gas of identical H-like atom having integer atomic no. has some atoms in the lowest (ground) energy level A and some atoms in a particular upper (excited) energy level B and there are no atoms in any other energy level. The atoms of the gas make transition to a higher energy level by absorbing monochromatic light of photon energy 8.7 eV. Subsequently, the atoms emit radiation of only ten different photons energies. Some of the emitted photons have energy 8.7 eV. Some have more and some have less than 8.7 eV.

3. Find the principal quantum number of initially excited level B.
(A) 2 (B) 3 (C) 4 (D) none of these
4. Find the ionisation energy for the gas atoms.
(A) -122.4 eV (B) 122.4 eV (C) -54.4 eV (D) 54.4 eV

Space for Rough Work

MATHEMATICS

SEC-1 : (Maximum Marks: 21)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

- The normals at A & B on $y^2 = 4x$ meet again on parabola at (p, q) , then $(AB)^2$ is -
 (A) $p^2 + 4p + 32$ (B) $p^2 + 4p - 32$ (C) $p^2 - 4p + 32$ (D) $p^2 - 4p - 32$
- A light ray gets reflected from the line $x = -2$. If the reflected ray touches the circle $x^2 + y^2 = 4$ and point of incident is $(-2, -4)$, then equation of incident ray is
 (A) $3x + 4y + 22 = 0$ (B) $4x + 3y + 20 = 0$
 (C) $x + 2y + 10 = 0$ (D) $x + y + 6 = 0$
- Consider lines $L_1 \equiv 2x - y + 10 = 0$, $L_2 \equiv 2x + y - 10 = 0$, $L_3 \equiv x - 15 = 0$ & $L_4 \equiv y - 5 = 0$. Let distances of point P from L_1, L_2, L_3 & L_4 be d_1, d_2, d_3 & d_4 respectively. If $d_1 = d_2$ and $d_3 = d_4$, then number of such possible points P in x-y plane is -
 (A) 0 (B) 2
 (C) 4 (D) infinitely many

Space for Rough Work

4. Locus of centroid of the triangle whose vertices are $(a \sin t, b \cos t)$, $(b \sin t, -a \cos t)$ and $(1, 0)$ is (where t is a parameter)
- (A) $a(3x - 1)^2 + b(3y)^2 = (a^2 - b^2)$ (B) $\frac{(3x - 1)^2}{(a + b)^2} + \frac{(3y)^2}{(a - b)^2} = 1$
- (C) $(3x + 1)^2 + (3y)^2 = (a^2 + b^2)$ (D) $(3x + 1)^2 + (3y)^2 = (a^2 - b^2)$
5. Suppose a, b denote the distinct real roots of the quadratic polynomial $x^2 + 20x - 2020$ and suppose c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 20x + 2020$. Then the value of $ac(a - c) + ad(a - d) + bc(b - c) + bd(b - d)$ is
- (A) 0 (B) 8000 (C) 8080 (D) 16000
6. Let a, b, c be the three numbers such that $(a - b)^2, (b - c)^2, (c - a)^2$ are in A.P. Then $a - b, b - c, c - a$ are always in $(a \neq b \neq c)$
- (A) A.P. (B) G.P. (C) H.P. (D) A.G.P.
7. If the equation $\left(\frac{x^2 + 5}{2}\right) = x - 2 \cos(ax + b)$ has atleast one solution, then $(a + b)$ can be equal to
- (A) 0 (B) π (C) 2π (D) 4π

Space for Rough Work

SEC-2 : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. In a triangle PQR, P is the largest angle and $\cos P = \frac{1}{4}$. Further the incircle of the triangle touches the sides PQ, QR and RP at N, L and M respectively, such that the lengths of PN, QL and RM are consecutive odd integers. Then possible length(s) of the side(s) of the triangle is (are) :
 (A) 16 (B) 12 (C) 24 (D) 22
2. In a right angle triangle the hypoteneous is $2\sqrt{2}$ times the perpendicular drawn from opposite vertex. If other angles are α and β then
 (A) $\sin \alpha \sin \beta = \frac{1}{2\sqrt{2}}$ (B) $\sin \alpha \cos \beta = \frac{\sqrt{2}-1}{2\sqrt{2}}$
 (C) $\cos \alpha \cos \beta = \frac{1}{2\sqrt{2}}$ (D) $\sin \alpha \cos \beta = \frac{\sqrt{2}+1}{2\sqrt{2}}$

Space for Rough Work

3. If $0 < x < 1$, then
- (A) $\log_e \pi > \log_x e > \log_x \pi$ (B) $\log_x \pi > \log_e \pi > \log_x e$
- (C) $\log_x \frac{1}{\pi} > \log_{1/x} e > \log_x \pi$ (D) $|\log_x \pi| > |\log_x e| > \log_e \frac{1}{\pi}$
4. Tangents TA and TB are drawn to the circle $S \equiv x^2 + y^2 - 2y - 3 = 0$ from the point $T(3, 4)$. Which of the following statement(s) is(are) correct?
- (A) The power of point $T(3, 4)$ with respect to circle $S = 0$ is 14
- (B) The angle between tangents from $T(3, 4)$ to the circle $S = 0$ is $\frac{\pi}{3}$
- (C) The equation of circumcircle of ΔTAB is $x^2 + y^2 - 3x - 5y + 4 = 0$
- (D) The area of quadrilateral TACB is $3\sqrt{7}$ (square units), where C is the centre of circle $S = 0$

Space for Rough Work

5. System of linear equations in x, y, z
 $2x + y + z = 1$
 $x - 2y + z = 2$
 $3x - y + 2z = 3$ have infinite solutions which
- (A) can be written as $(-3\lambda - 1, \lambda, 5\lambda + 3) \forall \lambda \in \mathbb{R}$
 (B) can be written as $(3\lambda - 1, -\lambda, -5\lambda + 3) \forall \lambda \in \mathbb{R}$
 (C) are such that every solution satisfy $x - 3y + 1 = 0$
 (D) are such that none of them satisfy $5x + 3z = 1$
6. If $A = \sum_{r=7}^{2400} \log_7 \left(\frac{r+1}{r} \right)$; $B = \prod_{r=2}^{1023} \log_r(r+1)$; $C = \sum_{r=2}^{2011} \frac{1}{\log_r 2 + \log_r 3 + \dots + \log_r 2011}$ then which of the following statements is/are true?
- (A) $3A + C = B$ (B) $A + B + C = 0$ (C) $A + C = B$ (D) $B - 2A = 4C$
7. Consider the series 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 3, 3, 3, 4..... such that value r repeats upto r^2 times, then,
- (A) sum of 1000 terms will be 10815 (B) sum of 1000 terms will be 14400
 (C) 97th term is equal to 98th term (D) 97th term is not equal to 98th term

Space for Rough Work

SEC-3 : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Consider the following lines

$$L_1 \equiv x - y - 1 = 0$$

$$L_2 \equiv x + y - 5 = 0$$

$$L_3 \equiv y - 4 = 0$$

Line L_1 is axis to the parabola, L_2 is tangent at vertex to this parabola & line L_3 is another tangent to this parabola at some point P. Let 'C' be the circle circumscribing the triangle formed by tangent and normal at point 'P' and axis of parabola.

On the basis of above information, answer the following :

1. The equation of the circle 'C' is -

(A) $x^2 + y^2 - 2x - 31 = 0$

(B) $x^2 + y^2 - 2y - 31 = 0$

(C) $x^2 + y^2 - 2x - 2y - 31 = 0$

(D) $2x^2 + 2y^2 - x - y - 62 = 0$

2. The given parabola is equal to which of the following parabolas -

(A) $y^2 = 16\sqrt{2}x$

(B) $x^2 = -4\sqrt{2}y$

(C) $y^2 = -\sqrt{2}x$

(D) $y^2 = 8\sqrt{2}x$

Space for Rough Work

Paragraph for Questions 3 and 4

Given two straight lines $\ell_1 : 3x + 4y = 5$ and $\ell_2 : 4x - 3y = 15$.

The equation of straight lines through $(1, 2)$ is $\ell_3 : x + ay + b = 0$ and $\ell_4 : px + y + q = 0$ such that triangle formed by ℓ_1, ℓ_2, ℓ_3 and ℓ_1, ℓ_2, ℓ_4 is isosceles triangle.

3. The value of $a + b + p + q$ is -
 (A) 1 (B) 2 (C) 3 (D) 4
4. Difference of area formed by lines ℓ_1, ℓ_2, ℓ_3 and ℓ_1, ℓ_2, ℓ_4 is $\frac{\lambda}{\delta}$, then $\lambda + \delta$ is (λ and δ are coprime)
 (A) 229 (B) 455 (C) 127 (D) 152

Space for Rough Work

Space for Rough Work